

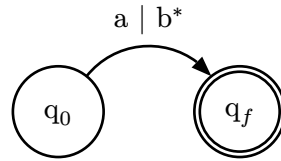
1.

$(([0 - 9]^*.[0 - 9]^+([eE]([-+] \mid \emptyset)[0 - 9]^+ \mid \emptyset) \mid [0 - 9]^+[eE]([-+] \mid \emptyset)[0 - 9]^+) \mid$
 $((0x|0X)(([0 - 9 \text{ a} - \text{f A} - \text{F}]^*.[0 - 9 \text{ a} - \text{f A} - \text{F}]^+ \mid [0 - 9 \text{ a} - \text{f A} - \text{F}]^+.) \mid [0 - 9 \text{ a} - \text{f A} - \text{F}]^+)$
 $[pP]([-+] \mid \emptyset)[0 - 9]^+))$
 $([fFL] \mid \emptyset)$

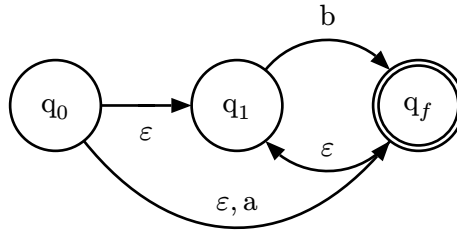
Note: We use the extension for character classes, but “.” matches a literal “.”. Sorry that it’s so long. Maybe it’s easier to read the typst code ;).

2.

With this rule, the NFSM



would expand to



But this also accepts “abbbb”.

3.

$$(r_1 \mid \emptyset) = R_1 \cup \emptyset = R_1$$

$$(r_1 \varepsilon) = \{x.y \mid x \in R_1, y \in \{\varepsilon\}\} = \{x.\varepsilon \mid x \in R_1\} = \{x \mid x \in R_1\} = R_1$$

4.

$X = \{\varepsilon\}$ for all L :

$$X = L^* \setminus L^+ = L^* \setminus (L^* \setminus L^0) = L^0 = \{\varepsilon\}$$